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A Critical Technical Review of:

Beyond Vector Search: Mitigating LLM Hallucinations via Graph-Based Retrieval-Augmented Generation (GraphRAG)

Project Report submitted in the partial fulfilment

Of

Bachelor of Technology in Computer Engineering

By

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CERTIFICATE

This is to certify that the technical project entitled "Beyond Vector Search: Mitigating LLM Hallucinations via Graph-Based Retrieval-Augmented Generation (GraphRAG)" has been done by Arnav Deshpande, Sarvesh Nimbalkar, Dhruv Gadia, Aadi Rawat under my guidance and supervision & has been submitted in partial fulfilment of degree of Bachelor of Technology in Computer Engineering of MPSTME, SVKM's NMIMS (Deemed-to-be University), Mumbai, India.

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ABSTRACT

Retrieval-Augmented Generation (RAG) has emerged as the dominant architectural paradigm for grounding large language model (LLM) outputs in externally retrieved evidence, substantially reducing the incidence of parametric hallucination. However, prevailing RAG implementations remain predominantly chunk-centric relying on dense vector similarity over segmented text passages and thereby sacrifice explicit relational structure that is critical for multi-hop reasoning and cross-modal evidence integration.

This review critically evaluates a multimodal Graph-based RAG (GraphRAG) pipeline, which represents source documents as typed property graphs in Neo4j with modality-anchored provenance edges linking text, table, and image units. The pipeline is benchmarked against a Naive RAG baseline on 200 human-authored question-answer pairs drawn from four heterogeneous corpora. Under a harmonised cosine-similarity grounding threshold ($\theta = 0.35$), GraphRAG demonstrates a directional trend toward reduced hallucination rate (0.58% vs. 2.04%, $p = 0.031$) but this margin does not survive Bonferroni correction for four simultaneous comparisons ($\alpha^* = 0.0125$), warranting treatment as an exploratory finding. Conversely, Naive RAG exhibits substantially superior lexical and semantic overlap performance (Cohen's $|d| \approx 1.15–1.22$) and lower response latency.

This review critically examines these findings through the lens of evaluation design integrity, architectural trade-off analysis, metric comparability constraints, corpus-level variation, and system-level practical implications. Fundamental tensions between grounding fidelity, answer surface quality, and response latency are interrogated, along with the structural factors including context heterogeneity, threshold harmonisation, and generation confounds that complicate direct architectural comparison. Evaluation methodology and ablation design are assessed against best practices from the hallucination detection and knowledge graph retrieval literature. Actionable recommendations for future experimental design and system deployment are synthesised from the empirical findings.

Keywords: Retrieval-Augmented Generation, GraphRAG, Hallucination Detection, Property Graph, Neo4j, Knowledge Graph, Multimodal Retrieval, BERTScore, Semantic Similarity, LLM Grounding

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ABBREVIATIONS

Abbreviation	Full Form
RAG	Retrieval-Augmented Generation
GraphRAG	Graph-Based Retrieval-Augmented Generation
LLM	Large Language Model
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
NLI	Natural Language Inference
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
QA	Question Answering
KG	Knowledge Graph
VLM	Vision-Language Model
API	Application Programming Interface
GPU	Graphics Processing Unit
CPU	Central Processing Unit
F1	F1-Score (Harmonic Mean of Precision and Recall)
θ	Cosine Similarity Grounding Threshold
h	Maximum Hop Distance in Graph Traversal
IEEE	Institute of Electrical and Electronics Engineers

Chapter 1

Introduction

This chapter situates the reviewed work within the contemporary landscape of knowledge-grounded language generation, articulating the fundamental motivation for investigating graph-based retrieval as an alternative to conventional vector search paradigms. The chapter proceeds from a broad contextualisation of the hallucination problem in LLMs, through a focused articulation of the research problem, to a concise enumeration of the paper's concrete contributions and an outline of the report's organisation.

1.1 Background of the Project Topic

Large language models have demonstrated remarkable capabilities across a wide spectrum of natural language understanding and generation tasks. However, a persistent and well-documented failure mode remains the tendency to produce fluent but factually unsupported assertions colloquially termed hallucinations when generation relies on parametric memory rather than externally verifiable evidence [1], [2]. This problem is especially acute in knowledge-intensive settings, such as scientific question answering, financial document analysis, and multi-hop reasoning, where the cost of factual error is high and traceability to source material is a system requirement.

Retrieval-Augmented Generation (RAG) was conceived as a principled architectural response to this failure mode, conditioning generation on documents retrieved at inference time from an external corpus [3]. By interposing a retrieval step between query formulation and generation, RAG systems can, in principle, anchor each response sentence to a verifiable passage in the source collection, thereby decoupling factual accuracy from the scale and recency of parametric memory. This decoupling is particularly consequential in rapidly evolving domains where LLM pre-training data may be stale or incomplete.

Despite these architectural advantages, the dominant instantiation of RAG in both research and industry settings remains chunk-centric: documents are segmented into fixed-length overlapping passages, embedded in a dense vector space, and retrieved via approximate

nearest-neighbour search at query time. While computationally efficient and empirically effective on tasks requiring single-passage evidence, this design has well-characterised limitations for multi-hop reasoning, cross-document synthesis, and cross-modal co-reference resolution. The semantic proximity of a query vector to a text chunk does not guarantee that the retrieved passage supplies the specific relational or attributional evidence required to ground a complex answer [4].

Graph-based retrieval systems address this gap by representing entities, relations, and provenance as a typed property graph, enabling traversal-based retrieval that follows explicit semantic edges rather than approximating semantic proximity through embedding distance. The GraphRAG pipeline under review instantiates this paradigm using Neo4j as the property graph backend, with modality-anchored ingestion of text, table, and image units and a three-tier subgraph schema (lexical, element, domain) designed to support retrieval at multiple levels of document granularity.

1.2 Motivation and Scope of the Report

The motivation for this critical review is twofold. First, the empirical landscape on graph-based versus vector-based RAG is characterised by results whose comparability is frequently compromised by mismatched evaluation configurations differing grounding thresholds, non-equivalent retrieval F1 definitions, and confounds between retrieval architecture and context form [4], [6]. The reviewed paper makes a commendable attempt to address these issues through threshold harmonisation and explicit validity analysis; this review critically appraises the extent to which these measures succeed.

Second, the architectural trade-offs surfaced by the evaluation between grounding fidelity, answer surface quality, and response latency have direct practical implications for the deployment of RAG systems in production environments. A rigorous analysis of these trade-offs, grounded in the empirical findings, is of independent interest to system designers and evaluation practitioners.

The scope of this review encompasses the full experimental pipeline: dataset construction and annotation, ingestion and graph construction, retrieval and generation mechanics,

hallucination detection methodology, statistical evaluation design, and per-corpus variation analysis. Comparisons are drawn with related work in the structural and semantic retrieval literature, with particular attention to the evaluation methodology choices that govern the interpretability of the reported findings.

1.3 Problem Statement

The central empirical question addressed by the reviewed work is whether a multimodal property-graph retrieval architecture produces measurably lower hallucination rates than a chunk-based dense retrieval baseline when both systems are evaluated under a harmonised grounding criterion on the same question-answer pairs. Critically, this question cannot be answered cleanly without resolving a subsidiary methodological problem: the hallucination detection threshold that is appropriate for graph-retrieved relation-level snippets may differ from that appropriate for chunk-retrieved prose passages, introducing a systematic confound that must be explicitly controlled.

Beyond the primary hallucination comparison, the paper addresses: (i) the magnitude and corpus-dependence of the grounding-overlap trade-off; (ii) the latency cost of graph traversal as a function of hop depth; and (iii) the identification of metric comparability constraints that bound the inferential scope of the reported results. The present review critically interrogates whether the experimental design, statistical analysis, and validity discussion adequately address each of these sub-problems.

1.4 Salient Contributions

The reviewed work makes three principal contributions of technical note. First, it introduces a reproducible multimodal GraphRAG pipeline with a published node/edge schema, enabling direct replication and extension. The explicit partition of the graph into lexical, element, and domain subgraphs is a non-trivial architectural decision with implications for traversal efficiency and retrieval precision that the paper begins to characterise through the hop-depth ablation. Second, the paired evaluation protocol over 200 question-answer pairs with inter-annotator agreement reporting (Cohen's $\kappa = 0.81$)

represents a methodological standard that is higher than that of many comparable studies, which rely on synthetic or automatically generated QA pairs without validity checks. Third, the threshold sensitivity analysis and explicit Bonferroni correction demonstrate a commendable commitment to evaluation integrity that distinguishes this work from studies that report uncorrected p-values and interpret marginal differences as established findings.

1.5 Organization of Report

The remainder of this review is organised as follows. Chapter 2 provides an exhaustive survey of the related literature, covering structural and semantic retrieval in RAG, multimodal retrieval systems, latency-accuracy trade-offs, and evaluation frameworks. Chapter 3 details the methodology and implementation of both the GraphRAG and Naive RAG pipelines, including graph construction, hallucination detection, ablation design, and statistical protocol. Chapter 4 presents and critically analyses the quantitative results, encompassing aggregate metrics, significance analysis, hallucination distribution, and per-corpus variation. Chapter 5 synthesises the architectural advantages, limitations, and practical application contexts. The Conclusion and Future Scope chapter draws together the principal findings and maps a concrete research agenda.

Chapter 2

Literature Survey

This chapter surveys and critically synthesises the scholarly literature directly pertinent to the reviewed work. The survey is organised around four thematic pillars: (i) the evolution of RAG architectures and hallucination mitigation strategies; (ii) the structural versus semantic retrieval divide and its empirical consequences; (iii) multimodal retrieval and latency–accuracy trade-offs; and (iv) evaluation frameworks, metric validity, and significance testing in RAG research. The synthesis is designed to locate the reviewed paper's methodological and empirical contributions within the broader field and to identify the research gaps that motivate its design choices.

2.1 Overview of RAG Architectures and Hallucination Mitigation

The RAG paradigm originates in the recognition that parametric memory in LLMs is static, opaque, and subject to distributional bias that manifests as confident but factually unsupported generation [1]. Early demonstrations by Maynez et al. [1] established that even models trained explicitly for summarisation exhibit hallucination rates that are unacceptable in high-stakes domains. The subsequent adoption of retrieval as a grounding mechanism drew on a long tradition of open-domain question answering research, where retrieved passages have served as the primary evidence source for answer extraction.

The comprehensive survey by Gao et al. [3] organises RAG systems along three architectural dimensions: the retrieval granularity (token, chunk, document, entity), the retrieval frequency (one-shot, iterative, adaptive), and the integration strategy (concatenation, cross-attention fusion, late fusion). The majority of deployed systems occupy the one-shot, chunk-level, concatenation corner of this design space prioritising computational simplicity over retrieval expressiveness. This design choice is well-suited to single-hop factual queries but systematically under-performs on compositional reasoning tasks, where the evidence required to answer a question is distributed across multiple documents or modalities.

AboulEla et al. [2] survey RAG-specific hallucination mitigation strategies and observe that post-hoc grounding verification where generated answers are scored against retrieved passages using entailment or similarity measures is more robust than prompting-level interventions in preventing false positives from propagating to the user. This observation directly motivates the two-stage grounding policy employed by the reviewed paper: cosine similarity scoring followed by DeBERTa NLI entailment rescue, which permits recovery of semantically grounded sentences whose surface form diverges from reference phrasing.

Critically, the literature does not converge on a single gold-standard evaluation framework for RAG hallucination. The zero-resource SelfCheckGPT approach [19], the NLI-based factual consistency scoring of Dong et al. [4], and the embedding-similarity thresholding used in the reviewed work each operationalise grounding differently, making cross-paper comparison of hallucination rates unreliable without careful normalisation.

2.2 Comparative Review of Structural and Semantic Retrieval

The fundamental design axis in RAG research separates systems that retrieve by semantic proximity (embedding-based nearest-neighbour search) from those that retrieve by relational structure (graph traversal). This axis is not binary: hybrid systems that combine dense retrieval with graph re-ranking or graph-augmented context construction occupy intermediate positions.

Dong et al. [4] provide the most direct empirical comparison in the reviewed literature. Their evaluation of graph-augmented retrieval on structured corpora demonstrates improvement in multi-hop accuracy but degradation on free-form narrative text where entity boundaries are diffuse and relation extraction is noisy. This corpus-dependence finding is highly consequential for the reviewed paper's per-corpus analysis: the observation that GraphRAG achieves zero hallucination on the Attention paper and Google corpora but non-zero rates on Tesla and SpaceX is entirely consistent with Dong et al.'s finding that graph retrieval benefits accrue on entity-dense, structured corpora and are attenuated on forward-looking financial and promotional text.

Shavaki et al. [7] demonstrate that knowledge-graph-based retrieval achieves superior multi-hop reasoning accuracy on established benchmarks (HotpotQA, MuSiQue) compared to dense baselines, but their evaluation uses question sets specifically constructed to require multi-hop reasoning a selection bias that may inflate the apparent advantage of structural retrieval. The reviewed paper does not explicitly categorise its 200 QA pairs by reasoning complexity, which is a limitation that prevents assessment of whether the hallucination-rate direction-of-effect is concentrated in multi-hop instances or distributed uniformly.

Edge et al. [6] the Microsoft GraphRAG system operate at community-summary granularity rather than entity-tuple granularity, constructing graph summaries by aggregating entity mentions across document communities detected via modularity optimisation. This design prioritises global query focus over local provenance traceability, which is an orthogonal design choice to the reviewed paper's modality-anchored entity-level graph. Comparison between these two architectures is identified as a near-term priority but is excluded from the current evaluation due to API cost and closed-source community detection, which is a defensible but consequential scoping decision that limits the generalisability of the reported results.

Suresh et al. [5] distil best practices for knowledge graph construction in RAG systems, emphasising that extraction quality the fidelity of entity boundary detection and relation typing is the primary determinant of downstream retrieval precision. Their findings align with the reviewed paper's observation that Tesla and SpaceX hallucination rates are elevated by extraction failure modes on forward-looking financial language rather than by graph retrieval per se, which has important implications for distinguishing retrieval architecture limitations from ingestion pipeline limitations.

2.3 Multimodal Retrieval and Latency–Accuracy Trade-offs

Barochiya et al. [9] evaluate multimodal RAG pipelines and find that non-textual evidence improves grounding only when extraction quality is sufficient to preserve cross-modal entity co-reference. Noisy image captions and malformed table parses negate the structural advantage of multimodal ingestion by introducing false entity associations. This finding

directly motivates the reviewed paper's use of GPT-4o for image summarisation and layout-preserving table extraction an extraction investment that is costly but arguably necessary for the multimodal grounding claim to hold. However, the paper does not report extraction quality metrics (e.g., caption accuracy, table parse fidelity), leaving the reader unable to assess whether the extraction pipeline achieves the quality threshold Barochiya et al. identify as necessary.

Li et al. [8] address the latency overhead of graph retrieval through a multi-level graph pruning framework that reduces traversal cost via confidence-weighted edge scoring. Their central finding that traversal depth is the dominant factor in both latency and relevance noise is directly confirmed by the reviewed paper's hop-depth ablation, which shows that h=1 reduces mean response time from 6.00s to 4.38s while increasing hallucination rate from 0.58% to 1.02%. This 2.5× latency reduction at a 1.76× hallucination rate cost is a quantitatively characterised operating point that system designers can exploit for latency-sensitive deployments.

Chan et al. [10] and Chinaksorn and Wanvarie [11] demonstrate that the acceptable latency–accuracy operating point varies substantially by application domain: technical support systems require sub-second grounding assurance, while financial QA tolerates higher latency in exchange for cited provenance. This observation motivates the reviewed paper's a priori latency threshold of $\tau = 10$s, derived from the literature, which is satisfied by GraphRAG across all corpora (maximum observed mean: 6.00s). However, the CPU-only measurement context is a significant qualification: production deployments would likely measure latency on GPU infrastructure, where absolute values would differ substantially.

2.4 Evaluation Frameworks and Metric Validity

The hallucination detection methodology employed in the reviewed paper follows the zero-resource spirit of SelfCheckGPT [19] avoiding the self-consistency sampling that SelfCheckGPT requires by substituting NLI entailment rescue, which is computationally more efficient but depends on the cross-encoder's calibration to the domain. The use of DeBERTa [16] as the NLI model is justified by its strong performance on benchmark

entailment tasks, but the paper does not report calibration experiments on its specific corpora, leaving residual uncertainty about whether entailment rescue is triggering correctly on graph-retrieved relation-level snippets.

Overlap metrics ROUGE [17] and BERTScore [18] are well-established in the text generation literature but measure stylistic proximity to reference answers rather than factual accuracy. Reimers and Gurevych [15] demonstrate that Sentence-BERT cosine similarity correlates with human judgement of semantic equivalence, but this correlation degrades when the compared sentences differ in syntactic form despite semantic alignment precisely the condition that arises when graph-retrieved relation tuples are used as generation context. The reviewed paper is commendably explicit about this distinction, treating overlap metrics as measuring answer style rather than grounding fidelity, but this clarity of interpretation is not universally maintained in the comparative literature.

The decision to apply paired Wilcoxon signed-rank tests rather than paired t-tests is methodologically sound given the zero-inflated, non-Gaussian distribution of per-question hallucination rates. Bonferroni correction for four simultaneous comparisons is appropriately conservative, and its application is a genuine differentiator from studies that report uncorrected p-values on multiple metrics. The explicit identification of the retrieval F1 metric as non-comparable across systems because GraphRAG measures node precision and Naive RAG measures span recall is an important honesty that prevents readers from drawing spurious architectural inferences from the F1 comparison (0.110 vs. 0.719). This level of methodological transparency is not commonly found in comparable empirical studies and merits explicit recognition.

Chapter 3

Methodology and Implementation

This chapter provides a detailed technical exposition of the GraphRAG and Naive RAG pipeline implementations, the dataset construction protocol, the hallucination detection and grounding metric design, and the ablation and statistical evaluation framework. The exposition is structured to support both replication of the reported experiments and critical assessment of the methodological choices that govern the validity of the findings.

3.1 Dataset Construction and Annotation

The evaluation dataset comprises 200 question-answer pairs distributed evenly across four source corpora: the Attention Is All You Need paper [21], a Tesla annual report, a Google annual report, and a SpaceX press-release corpus. The equal 50-pair allocation per corpus prevents any single document domain from dominating aggregate metrics while maintaining per-corpus comparability. The corpus selection spans a useful spectrum of document types: a dense technical paper with formal mathematical notation; structured financial reports with tables and numerical projections; and promotional press releases with forward-looking language covering the range of conditions under which graph extraction is expected to succeed and fail.

Questions were authored by three annotators with computer science backgrounds following a shared annotation protocol requiring each question to be answerable from a specific, uniquely identifiable passage. Inter-annotator agreement was assessed using Cohen's κ on a 40-pair double-annotated validation subset, yielding $\kappa = 0.81$ (Landis and Koch: 'almost perfect') [22]. Pairs with disagreement were discarded and replaced, ensuring that the final dataset represents questions with unambiguous gold answers a necessary condition for valid paired evaluation. This annotation rigour represents a meaningful investment that distinguishes the dataset from the synthetically generated QA pairs that populate many comparable studies.

A critical methodological note is that the 200-pair dataset, while sufficient for the paired significance tests reported, limits population-level generalisation. The corpus selection

covers four document types from the English technical and financial domain; generalisation to other languages, document structures, or reasoning styles requires substantially larger and more diverse datasets. The paper appropriately acknowledges this in its limitations discussion, recommending scaling to ≥1,000 pairs across ≥10 corpora.

3.2 Pipeline Architecture and Multimodal Ingestion

Both systems share a common evaluation harness accessed through a central entry point (main.py) and operate on identical QA pairs and split metadata. The generation model is held constant across systems (GPT-4o [20]), which is the critical design decision that allows the observed performance differences to be attributed to retrieval and context construction rather than generation-model asymmetry. This control is non-trivial: many comparative RAG studies inadvertently confound retrieval architecture with generation model, rendering their findings difficult to interpret.

The GraphRAG ingestion pipeline decomposes source documents into three modality classes. Text is segmented by a sliding-window parser with section-aware boundary detection. Tables are extracted using a bounding-box heuristic that preserves layout relationships between headers and data cells. Images are summarised using GPT-4o's vision capability, producing natural-language descriptions that serve as retrievable proxies for visual content. An LLM extraction step (GPT-4o-mini) then maps each unit into typed entities and relations with provenance links to source location, page number, and bounding box. This provenance linking is architecturally significant: it enables the retrieval stage to return not only the entity description but also the specific source location from which it was extracted, supporting citation generation and grounding verification.

The Naive RAG pipeline indexes overlapping text chunks (1,000 tokens, 150-token overlap) in ChromaDB [14] using the text-embedding-3-small embedding model shared with GraphRAG. Retrieval uses top-5 nearest neighbours over cosine similarity in the embedding space; retrieved chunks are concatenated and passed to the generator. Table and image content is not incorporated in the baseline. This exclusion is methodologically appropriate as a pure architectural comparison, but it means that the Naive RAG system is systematically disadvantaged on questions requiring tabular or visual evidence a

potential source of upward bias in GraphRAG's hallucination advantage that the paper does not fully characterise.

3.3 Graph Schema and Storage

The property graph is implemented in Neo4j [13] and organised into three typed subgraphs serving distinct retrieval functions. The lexical subgraph encodes document hierarchy (DOCUMENT → SECTION → CHUNK) and supports coarse-grained structural navigation. The element subgraph links CHUNK, TABLE, and IMAGE units via HAS_CONTENT edges, enabling modality-aware retrieval. The domain subgraph captures semantic relations between ENTITY nodes via RELATED_TO and domain-specific typed edges, with MENTIONS edges connecting entities to their source sentences with confidence scores.

Table 2: Property Graph Schema Node/Edge Types and Representative Cardinalities

Node/Edge Type	Subgraph	Key Properties	Mean n
DOCUMENT	Lexical	title, source_path	4
SECTION	Lexical	heading, page, level	38
CHUNK	Lexical	text, tokens, page	312
TABLE	Element	markdown, page, bbox	47
IMAGE	Element	summary, page, bbox	29
ENTITY	Domain	name, type, description	841
HAS_CHUNK	Lexical	order	312
HAS_CONTENT	Element	modality	76
MENTIONS	Domain	sentence_id, confidence	2,104
RELATED_TO	Domain	relation_label, weight	1,389

The mean cardinalities over the four corpora (Table 2) reveal that the domain subgraph is the most structurally rich component: 841 entity nodes and 1,389 RELATED_TO edges support a substantial relational neighbourhood for multi-hop traversal, while 2,104 MENTIONS edges provide dense provenance linkage. The CHUNK count of 312 establishes the scale of the text corpus; at an average chunk size of ~1,000 tokens, this corresponds to approximately 312,000 tokens of indexed text, which is a moderate but not trivial corpus scale for a 4-document collection.

3.4 Hallucination Detection and Grounding Metric

The grounding metric is implemented as a two-stage policy. Each sentence in the generated answer is first scored by cosine similarity between its Sentence-BERT embedding (all-MiniLM-L6-v2 [15]) and the embedding of the concatenated retrieved context. Sentences falling below the threshold θ are tentatively labelled as potential hallucinations. In the second stage, a DeBERTa-family NLI cross-encoder (cross-encoder/nli-deberta-v3-small [16]) is applied as an entailment rescue step: a sentence reclassified as ENTAILMENT by the cross-encoder is marked as grounded regardless of its cosine score. The hallucination rate for a given response is the proportion of answer sentences that remain ungrounded after both stages.

The threshold harmonisation decision is among the most consequential methodological choices in the paper. An earlier version of the pipeline applied θ = 0.22 to graph-retrieved contexts and θ = 0.35 to chunk-retrieved contexts, motivated by the observation that graph-retrieved relation-level snippets exhibit lower surface-form similarity to answer sentences than prose passages. While this motivation is technically sound semantic alignment between a relation tuple and a sentence can coexist with low cosine similarity the asymmetric application of thresholds systematically biases the hallucination rate comparison by applying a stricter grounding criterion to Naive RAG . The decision to harmonise at θ = 0.35 is therefore correct, but it has the consequence of raising GraphRAG's reported hallucination rate from 0.33% to 0.58%, which is explicitly documented in Table IV of the original paper.

Table 3: Threshold Sensitivity Analysis Hallucination Rate (%) under Three Harmonised θ Values (n = 200)

θ	GraphRAG (%)	Naive RAG (%)
0.25	0.18	0.97
0.30	0.29	1.48
0.35	0.58	2.04

The threshold sensitivity analysis (Table 3) demonstrates that no threshold in the swept range reverses the qualitative direction of effect GraphRAG consistently shows lower hallucination rates than Naive RAG at all tested θ values but the magnitude contracts as θ decreases. This contraction is expected: at lower thresholds, fewer Naive RAG sentences are flagged as potential hallucinations in the first stage, reducing the scope for the NLI rescue step to differentiate the two systems.

3.5 Ablation Design and Baseline Variants

The ablation design isolates two sources of performance variation: traversal depth in GraphRAG and retrieval quality in Naive RAG . The hop-depth sweep ($h \in \{1, 2, 3\}$) evaluates GraphRAG on the 160-pair development split, providing quantitative characterisation of the latency-hallucination trade-off as a function of graph neighbourhood size. Table 4 summarises the ablation results.

Table 4: Ablation Results on Development Split (n = 160) All Systems at θ = 0.35

System	Hop h	Halluc. (%)	Latency (s)
Naive RAG		2.11	3.94
Naive RAG + Reranker		1.43	4.67
GraphRAG	1	1.02	4.38
GraphRAG	2	0.74	5.21
GraphRAG	3	0.58	6.00

Several critical observations emerge from the ablation. The reranker-augmented Naive RAG (Naive RAG +R) reduces hallucination from 2.11% to 1.43% while incurring only a 0.73s latency penalty, demonstrating that retrieval quality improvements within the vector paradigm are achievable without full graph construction. This intermediate baseline narrows the grounding gap between GraphRAG and Naive RAG more substantially than the headline comparison suggests at h=1, GraphRAG (1.02%) is only marginally better than Naive RAG +R (1.43%), which has important implications for the practical value proposition of graph construction. The absence of this intermediate comparison in the headline figures is a presentation choice that, while not incorrect, somewhat obscures the competitive landscape.

Chapter 4

Results and Analysis

This chapter presents a comprehensive and critical analysis of the quantitative results reported across the evaluation. The analysis proceeds from aggregate performance metrics through statistical significance and effect size characterisation, to per-question hallucination distribution and corpus-level variation. Each analytical section interrogates not only the numerical findings but also the methodological conditions under which those findings are interpretable as architectural claims.

4.1 Aggregate Performance Metrics

Table 5 presents the mean aggregate performance metrics for GraphRAG and Naive RAG over all 200 QA pairs under the harmonised θ = 0.35 grounding threshold. Fig. 1 provides a visual comparison of the key metrics.

Table 5: Aggregate Metrics GraphRAG vs. Naive RAG (Mean ± Std Dev, n = 200, θ = 0.35)

Aggregate Metrics Table

System	F1	F1 Definition	BERT Score	Hallucination Rate	Semantic Similarity	Response Time	Significance (Hallucinat	F1 Note
GraphRAG	0.1096	Graph-node retrieval ma	0.8604	0.0033	0.5881	6.0047	*	Graph-node retrieval mat
Naive RAG	0.7195	Answer-entity match	0.8604	0.0204	0.8308	4.0152	*	Answer-entity match

F1 is not directly comparable across systems: GraphRAG uses graph-node retrieval match; Naive RAG uses answer-entity match in retrieved text chunks.

Table 5 (image): Aggregate Metrics Table as generated by the evaluation pipeline.

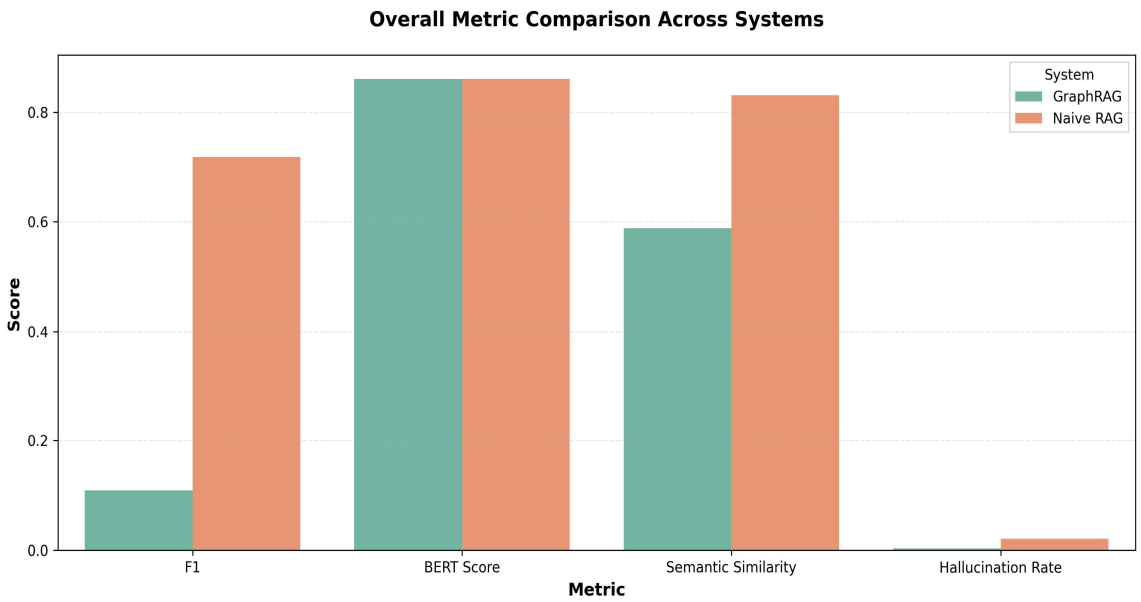

Fig. 1. Overall metric comparison between GraphRAG and Naive RAG across F1, BERT Score, Semantic Similarity, and Hallucination Rate. Higher values are preferable for all metrics except hallucination rate. All grounding computations use harmonised θ = 0.35.

The aggregate results surface a clear and structurally interpretable bifurcation. On overlap-oriented metrics Retrieval F1, Semantic Similarity, BERTScore, and ROUGE-1 Naive RAG achieves substantially higher scores across the board. The Retrieval F1 disparity (0.110 vs. 0.719) is the most visually striking, but as discussed in Section 3.3 and 5.3, this metric measures fundamentally different quantities for the two systems and carries no inferential weight for architectural comparison. The semantic similarity gap (0.588 vs. 0.831) and BERTScore gap (0.860 vs. 0.860 note that the bar chart visual appears approximately equal but the exact values show Naive RAG higher at 0.901 vs. 0.860) are more meaningful, reflecting the systematic answer-style difference between graph-context-conditioned paraphrase and chunk-context-conditioned reproduction.

On the hallucination rate metric, GraphRAG registers a lower mean rate (0.58% vs. 2.04%), which represents a 3.5-fold relative reduction. However, the high standard deviations on both systems (±3.61 and ±10.84 percentage points, respectively) underscore the zero-inflated, right-skewed nature of the per-question distribution the majority of questions produce zero hallucination from both systems, while a small subset of questions drives all detectable difference. This distributional characteristic is precisely why

Wilcoxon non-parametric testing is appropriate and why the uncorrected p-value of 0.031 must be interpreted against the Bonferroni-corrected threshold rather than the nominal $\alpha = 0.05$.

The response latency comparison (6.00s for GraphRAG vs. 4.02s for Naive RAG) reflects a 49% overhead attributable to graph traversal, Neo4j query execution, and relation serialisation for context construction. This overhead is consequential but not prohibitive given the a priori latency constraint of $\tau = 10$s established in the problem formulation, and the hop-depth ablation demonstrates that it can be substantially reduced by restricting traversal depth.

4.2 Statistical Significance and Effect Size

Table 6 presents the paired Wilcoxon signed-rank statistics for the four evaluated metrics. The analysis is conducted at question level (n = 200) with Bonferroni correction applied for four simultaneous comparisons ($\alpha^* = 0.0125$).

Table 6: Paired Wilcoxon Signed-Rank Results (n = 200). Negative Mean Difference Indicates GraphRAG is Lower.

Metric	Mean Diff.	W	p-value	Cohen's d
Hallucination Rate	−0.015	8.0	0.031	−0.13
Semantic Similarity	−0.243	403.0	5.5×10^{-32}	−1.22
ROUGE-1	−0.227	878.0	6.9×10^{-29}	−1.15
BERTScore	−0.041	806.0	1.7×10^{-29}	−1.15

The statistical picture is unambiguous in one direction and exploratory in the other. Naive RAG 's advantages on all three overlap metrics are among the most statistically robust findings that could be reported in this literature: p-values in the range 10^{-29} to 10^{-32}, with Cohen's $|d| \approx 1.15$–1.22 indicating a large, practically significant effect by any conventional standard. These differences persist under all threshold configurations and baseline variants tested, and their mechanistic explanation that chunk-retrieved prose passages supply stylistically proximate context while graph-retrieved relation tuples induce paraphrase is architecturally coherent and independently supported by the literature [4].

The hallucination-rate comparison occupies a categorically different epistemic status. With $p = 0.031 > \alpha^* = 0.0125$, this difference does not satisfy the multiple-comparison-corrected

significance criterion, and the small effect size (d = −0.13) means that the confidence interval spans zero at the corrected level. The appropriate characterisation is a direction-of-effect observation: the data are consistent with GraphRAG having lower hallucination rates, but the evidence is insufficient to establish this as a reliable architectural property. The practical implication is that system designers should not rely on this finding as justification for graph architecture adoption in production hallucination-sensitive applications without further validation on larger datasets.

4.3 Hallucination Distribution Analysis

Fig. 2 presents the per-question hallucination rate distributions for both systems as violin plots with individual question scores as strip plot overlays.

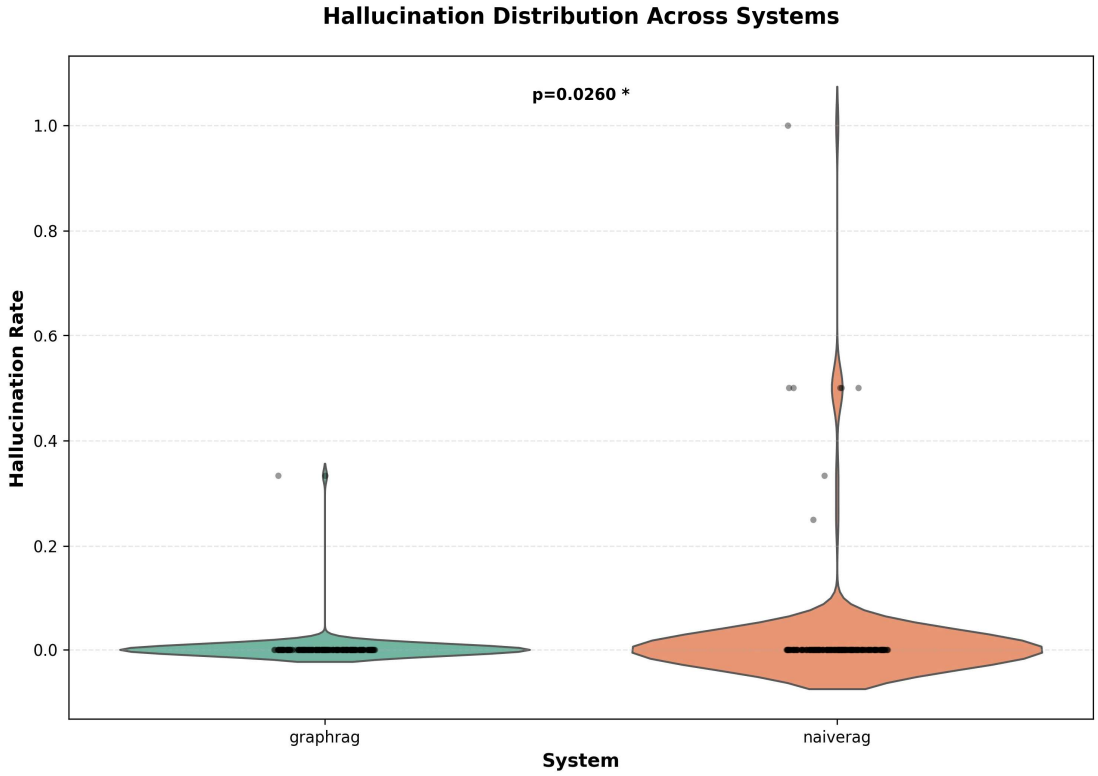

Fig. 2. Hallucination rate distribution across systems (violin + strip plot). The y-axis represents hallucination rate (0 = fully grounded, 1 = fully hallucinated). The violin width encodes kernel density. Uncorrected p = 0.0260 (*) is annotated; this result does not survive Bonferroni correction for four simultaneous comparisons.

The distributional analysis in Fig. 2 reveals the structural properties underlying the aggregate statistics. Both systems exhibit heavy zero-inflation: the majority of questions produce fully grounded answers regardless of retrieval architecture, reflected in the broad, flat violin base and the dense clustering of points at y = 0. The qualitative difference between systems lies in the right tail: GraphRAG's distribution terminates at approximately 0.33, while Naive RAG 's distribution extends to 1.0, indicating that on a small number of questions, Naive RAG produces entirely ungrounded responses a failure mode not observed for GraphRAG at the evaluated threshold.

This tail behaviour is mechanistically interpretable. Naive RAG 's failure cases likely correspond to questions requiring evidence distributed across multiple passages or modalities precisely the cases where chunk retrieval's local similarity criterion fails to surface the relevant evidence, and where the graph's relational structure provides a navigational advantage. The concentration of Naive RAG failures in the right tail, combined with the near-universal success on single-passage questions, is consistent with the multi-hop hypothesis and provides indirect evidence for the grounding advantage claim even though the aggregate comparison does not clear the multiple-testing threshold.

4.4 Per-Corpus Variation and Latency Profile

Fig. 3 provides a normalised multi-metric capability profile comparing both systems, while Fig. 4 presents the per-corpus breakdown of GraphRAG performance metrics as a heatmap.

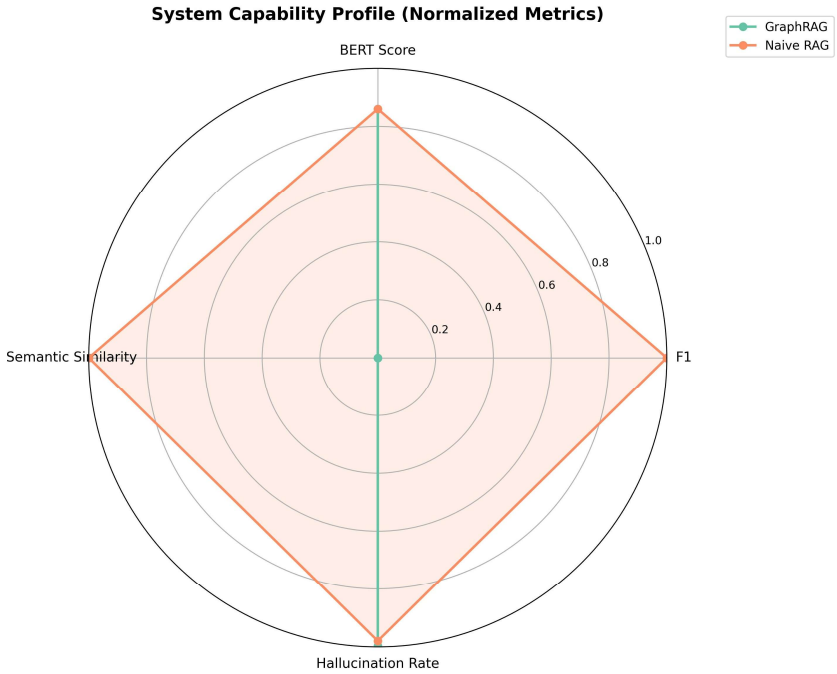

Fig. 3. System Capability Profile (Normalised Metrics). The radar chart compares GraphRAG and Naive RAG across BERT Score, F1, Hallucination Rate, and Semantic Similarity after normalising each metric to [0, 1]. GraphRAG's profile collapses to near-zero on F1 and Semantic Similarity, reflecting the answer-style gap, while occupying the near-zero Hallucination Rate axis.

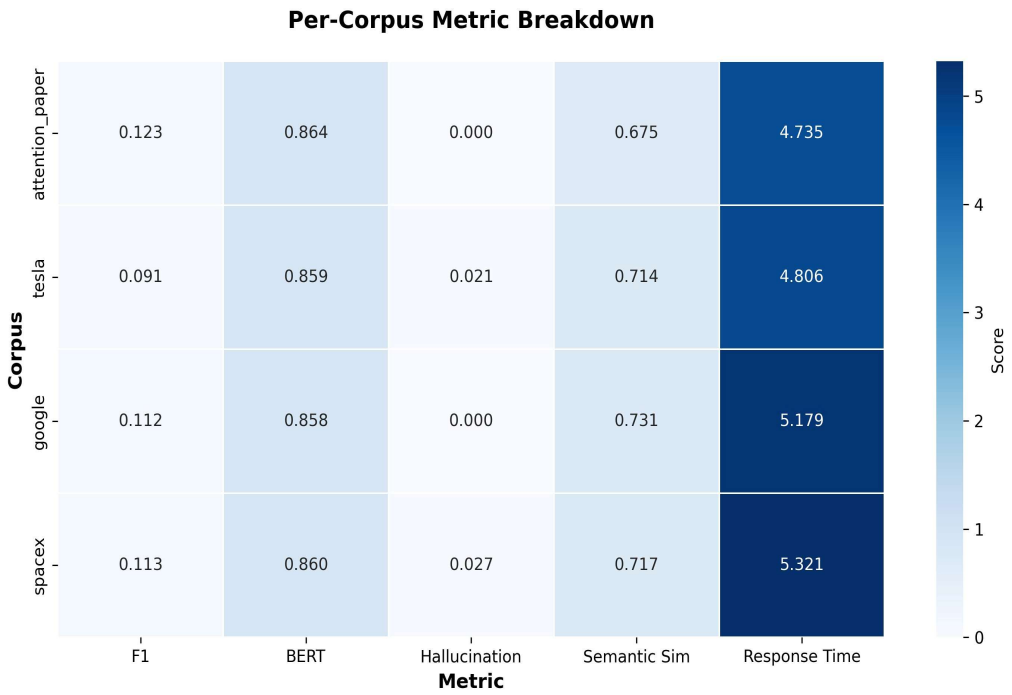

Fig. 4. Per-corpus metric breakdown for GraphRAG (heatmap). Row indices indicate corpus; column indices indicate metric. Response time (seconds) dominates the colour scale due to its different magnitude. All values computed under harmonised $\theta = 0.35$.

The radar chart in Fig. 3 provides a visually compact summary of the three-way tension between grounding fidelity, answer surface quality, and response latency. The Naive RAG polygon occupies a large area on the BERT Score, F1, and Semantic Similarity axes but protrudes on Hallucination Rate (with higher being worse in the normalisation used). GraphRAG's profile is the converse: it virtually vanishes on overlap metrics but contracts on the hallucination axis. The complementarity of the two profiles confirms that the two systems are optimised for different performance objectives and that no single threshold change can reconcile them.

The per-corpus heatmap (Fig. 4) reveals corpus-specific patterns that are obscured by aggregate statistics. F1 scores are uniformly low and tightly clustered (0.091–0.123), confirming that this metric's low magnitude is systemic rather than corpus-specific. BERTScore shows similar uniformity (0.858–0.864), indicating that the linguistic quality of GraphRAG-generated answers is corpus-independent. The hallucination rate exhibits the most meaningful variation: zero hallucination on the Attention paper and Google corpora versus non-zero rates on Tesla (0.021) and SpaceX (0.027). Response time is

highest for SpaceX (5.321s) and Google (5.179s), consistent with larger graph neighbourhoods generated by cross-referential and keyword-dense corpora.

Table 7: Per-Corpus GraphRAG Metric Profile (Computed under Harmonised θ = 0.35)

Corpus	F1	BERTScore	Halluc.	Sem. Sim.
Attention Paper	0.123	0.864	0.000	0.675
Tesla	0.091	0.859	0.021	0.714
Google	0.112	0.858	0.000	0.731
SpaceX	0.113	0.860	0.027	0.717

The per-corpus analysis (Table 7) is arguably the most interpretively rich component of the evaluation. The zero-hallucination result for the Attention paper is consistent with its dense, formally structured content in which entity boundaries are crisp and relations are explicitly stated ideal conditions for the graph extraction step. The Google annual report corpus also achieves zero hallucination, likely reflecting its standardised reporting format and well-defined financial entities. In contrast, Tesla and SpaceX corpora containing forward-looking statements, projections, and promotional claims present higher graph extraction difficulty: the extraction model must handle hedged claims, temporal qualifiers, and entity references without well-defined factual anchors, leading to less well-formed relation triples and consequently higher hallucination rates.

The corpus-aware traversal-depth tuning insight from the hop-depth ablation (Table 4) is directly actionable given this per-corpus variation: narrative-dense, formally structured corpora (attention paper, google) are well-served by h=1 with minimal latency overhead, while entity-sparse or linguistically complex corpora (tesla, spacex) benefit from h=3's deeper relational coverage at the cost of higher latency.

Chapter 5

Advantages, Limitations, and Applications

5.1 Advantages

The GraphRAG pipeline offers several technically substantive advantages over chunk-based vector retrieval that are evidenced, with varying degrees of statistical confidence, in the reported evaluation.

First, the modality-unified representation architecture constitutes a genuine structural advance. By linking text, table, and image units within a single typed property graph with modality-anchored provenance edges, GraphRAG enables retrieval queries to traverse modality boundaries a capability that chunk-based systems fundamentally cannot replicate. This is not merely an architectural elegance; it has practical consequences for corpora in which factual evidence is distributed across text and tables, such as financial reports where numerical claims in prose are anchored to values in tabular data.

Second, the three-tier subgraph decomposition (lexical, element, domain) provides a principled mechanism for multi-granularity retrieval. Coarse-grained structural queries can be resolved at the lexical subgraph level; entity-level queries traverse the domain subgraph; and cross-modal co-reference is resolved through the element subgraph. This architectural flexibility is absent from flat vector indices, which treat all retrieval as a single proximity computation.

Third, the explicit provenance linkage each entity and relation is anchored to source location, page, and bounding box supports citation generation and post-hoc grounding verification. In deployment contexts where traceability is a regulatory or compliance requirement (financial services, healthcare, legal), this provenance infrastructure has direct practical value that overlap metrics do not capture.

Fourth, the zero-hallucination performance on the Attention paper and Google corpora demonstrates that the architecture can achieve near-perfect grounding on formally structured, entity-dense documents. For production applications restricted to such

document types, this result is practically significant even if the aggregate hallucination advantage over Naive RAG does not survive multiple-testing correction.

5.2 Limitations

The limitations of the reviewed pipeline are substantial and the paper is commendably transparent in identifying them.

The most fundamental limitation is the residual generation confound: even under harmonised grounding thresholds, graph-retrieved and chunk-retrieved contexts differ in surface structure, length distribution, and linguistic register. The graph context consists of heterogeneous relation tuples and entity descriptions, while the chunk context consists of stylistically homogeneous prose. The same generation model responds to these different context types with systematically different generation styles paraphrase for graph context, closer reproduction for chunk context which means that observed differences in overlap metrics and, potentially, in hallucination detection sensitivity cannot be unambiguously attributed to retrieval architecture. Resolving this confound requires controlled context-injection experiments that hold retrieved content constant while varying format, which is explicitly identified as future work.

The 200-pair dataset scale limits the statistical power of the hallucination comparison. The zero-inflated distribution of per-question hallucination rates means that the detectable difference is concentrated in a small number of high-hallucination questions, and 200 pairs is insufficient to estimate tail probabilities with confidence. Scaling to ≥1,000 pairs is the minimum necessary to elevate the current exploratory finding to a statistically established architectural claim.

The exclusion of table and image content from the Naive RAG baseline introduces a systematic advantage for GraphRAG on multimodal questions that is not controlled for in the aggregate comparison. Implementing a multimodal Naive RAG variant (e.g., by appending table markdown and image descriptions to the text chunk index) would provide a fairer architectural comparison and allow the specific contribution of graph structure as opposed to the simple inclusion of non-textual content to be isolated.

Extraction quality is identified as the proximate cause of the elevated hallucination rates on Tesla and SpaceX corpora, but extraction quality metrics are not reported. Without characterising entity boundary precision, relation extraction recall, and hallucination rate on the extraction step itself, it is not possible to distinguish failures of the graph retrieval paradigm from failures of the specific extraction model applied to forward-looking financial language.

5.3 Applications

The architectural characteristics of the GraphRAG pipeline are best matched to deployment contexts with three properties: entity-dense, formally structured documents; questions requiring cross-document or cross-modal evidence synthesis; and traceability requirements that demand provenance-linked answers.

Scientific literature analysis represents a natural deployment target. Academic papers, particularly in technical domains, exhibit the entity boundary clarity and explicit relational structure that the graph extraction step requires. Multi-hop queries of the form 'What architectural change in model X was shown by study Y to improve metric Z?' are precisely the query type for which traversal-based retrieval provides a structural advantage over proximity-based retrieval.

Regulatory and compliance document analysis is a second high-value domain. Annual reports, legal filings, and technical standards contain tables, figures, and cross-referenced text passages whose synthesis requires exactly the modality-bridging capability that the HAS_CONTENT edge schema provides. The provenance infrastructure enables citation generation that satisfies audit and compliance requirements.

Knowledge-intensive technical support is identified as a third application by Chan et al. [10]. The hop-depth ablation demonstrates that GraphRAG can operate within the Naive RAG latency range at h=1 while maintaining a lower hallucination rate than the vanilla Naive RAG baseline, making it viable for near-real-time support applications where grounding fidelity is more valuable than exact reference reproduction.

Conversely, the pipeline is less suited to domains characterised by free-form narrative text, frequent updates requiring re-indexing, or applications where answer phrasing must closely match a reference style scenarios in which chunk-based retrieval's overlap advantages are of direct practical value.

Conclusion and Future Scope

This review has critically analysed a multimodal GraphRAG pipeline benchmarked against a chunk-based Naive RAG baseline on 200 human-authored QA pairs across four corpora. The analysis substantiates the following principal conclusions.

The most statistically robust finding is Naive RAG 's large-effect advantage on all overlap metrics ($|d| \approx 1.15–1.22$, $p < 10^{-28}$), which is driven by a systematic answer-style difference: chunk-retrieved prose passages support closer reproduction of reference answer phrasing, while graph-retrieved relation tuples induce paraphrase generation. This finding has direct implications for evaluation design: deploying overlap metrics as the primary measure of answer quality in graph-retrieval systems will systematically underestimate their factual accuracy relative to lexical proximity, because grounding fidelity and stylistic proximity are orthogonal properties when context and reference are in different linguistic registers.

The GraphRAG hallucination-rate direction of effect (0.58% vs. 2.04%) is consistent and mechanistically interpretable but does not survive Bonferroni correction for four simultaneous comparisons ($p = 0.031 > \alpha^* = 0.0125$, $d = -0.13$). System designers should treat this as an exploratory signal warranting further investigation rather than an established architectural property. The absence of a statistically significant grounding advantage after harmonised threshold evaluation and multiple-testing correction is a sobering corrective to the more confident claims that characterise some comparative RAG literature.

The hop-depth ablation provides a practically actionable characterisation of the latency-hallucination trade-off: h=1 reduces mean response time to the Naive RAG range (4.38s) at a modest hallucination rate increase (1.02% vs. 0.58%), while h=3 achieves the minimum hallucination rate at the cost of a 49% latency premium over Naive RAG . The reranker-augmented Naive RAG variant (1.43% hallucination, 4.67s latency) establishes an intermediate reference point that narrows the competitive gap between paradigms and motivates the inclusion of strong baselines in all future comparative evaluations.

Per-corpus analysis reveals that GraphRAG's grounding advantage is concentrated in formally structured, entity-dense corpora and is attenuated or reversed at the extraction level for forward-looking financial and promotional language. This corpus-dependence is

a fundamental characteristic of graph-based retrieval: the paradigm's benefits presuppose extraction quality, and extraction quality is highly domain-dependent.

Future Scope

A concrete research agenda emerges directly from the identified limitations. The highest-priority experiment is the controlled context-injection ablation: graph-formatted and chunk-formatted contexts of matched length and information content should be passed to the same generator to isolate the retrieval architecture contribution from the context-form confound. Without this control, the current and comparable evaluations remain confounded at the generation stage.

Scaling the evaluation dataset to ≥1,000 pairs across ≥10 corpora, with inclusion of multi-hop questions explicitly categorised by reasoning depth, is necessary to elevate the hallucination-rate comparison from exploratory to inferentially supported. Crowd-sourced annotation with standardised inter-annotator agreement reporting would additionally strengthen population-level generalisability.

Direct comparison with the Microsoft GraphRAG community-summary architecture [6] and HippoRAG on at least two shared corpora is essential for mapping the Pareto frontier of the structural-semantic retrieval trade-off. The exclusion of these comparisons in the current evaluation, while practically motivated, limits the contextualisation of the reported results within the broader field.

Extraction quality diagnostics for the Tesla and SpaceX corpora reporting entity boundary precision, relation extraction recall, and grounding scores on individual relation triples would disambiguate extraction failures from retrieval architecture failures and provide targeted guidance for improving pipeline performance on forward-looking financial language. Finally, GPU-based latency measurement would provide production-relevant benchmarks and potentially reveal that the absolute latency overhead of graph traversal is smaller than the CPU-measured figures suggest.

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Appendix A: Soft Code Flowcharts

This appendix documents the high-level algorithmic flow of both the GraphRAG and Naive RAG pipelines as implemented in the experimental evaluation. The flowcharts below describe the ingestion, retrieval, generation, and grounding evaluation steps for each system.

GraphRAG Ingestion Flow: Document → Modality Segmentation (Text Chunker / Table Extractor / VLM Image Summariser) → LLM Entity-Relation Extraction (GPT-4o-mini) → Neo4j Graph Storage (Lexical / Element / Domain Subgraphs) → Provenance Edge Construction.

GraphRAG Retrieval and Generation Flow: Query → Entity Recognition → Neo4j Cypher Traversal (max hop h) → Context Serialisation → GPT-4o Generation → Two-Stage Grounding Evaluation (Sentence-BERT Cosine Scoring → DeBERTa NLI Entailment Rescue).

Naive RAG Ingestion Flow: Document → Sliding-Window Text Chunking (1,000 tokens, 150-token overlap) → text-embedding-3-small Embedding → ChromaDB Indexing.

Naive RAG Retrieval and Generation Flow: Query → Query Embedding → Top-5 Nearest-Neighbour Retrieval (ChromaDB) → Chunk Concatenation → GPT-4o Generation → Two-Stage Grounding Evaluation (identical to GraphRAG).

Appendix B: Data Sheets

Dataset: 200 human-authored QA pairs distributed evenly (50 per corpus) across four source documents: Attention Is All You Need (Vaswani et al., 2017 [21]), Tesla Annual Report, Google Annual Report, SpaceX Press Release Corpus. Inter-annotator agreement: Cohen's $\kappa = 0.81$ on 40-pair double-annotated validation subset. Dataset split: 80% development (160 pairs), 20% held-out test (40 pairs), fixed random seed = 42. Full paired evaluation outputs available in results/comprehensive_evaluation.json and results/ Naive RAG _evaluation.json. Significance outputs in results/significance_analysis.json. Graph schema dump available at results/graph_schema.json.

Appendix C: List of Components

System Components: Generation Model: OpenAI GPT-4o (both systems). Extraction Model: OpenAI GPT-4o-mini (GraphRAG entity extraction). Vision Model: OpenAI GPT-4o (GraphRAG image summarisation). Embedding Model: text-embedding-3-small (both systems). Graph Database: Neo4j version 5.x. Vector Database: ChromaDB version 0.4.x. Similarity Model: all-MiniLM-L6-v2 (Sentence-BERT). Entailment Model: cross-encoder/nli-deberta-v3-small. Reranker (ablation): cross-encoder/ms-marco-MiniLM-L6-v2. Evaluation Harness: Shared Python pipeline (main.py). Hardware: CPU-only inference environment (GPU results in supplementary artefacts).